we say that “this thing happens before that one,” because without time, there is no concept of “before” or “after.” Space, the sole remaining remnant of what was once a unified spacetime, becomes a random, probabilistic froth, like soapsuds.

Before its rupture (that is, outside the singularity), spacetime is like a piece of wood impregnated with water. In this analogy, the wood represents space, the water represents time, and the two (wood and water; space and time) are tightly interwoven, unified. The singularity and the laws of quantum gravity that rule it are like a fire into which the water-impregnated wood is thrown. The fire boils the water out of the wood, leaving the wood alone and vulnerable; in the singularity, the laws of quantum gravity destroy time, leaving space alone and vulnerable. The fire then converts the wood into a froth of flakes and ashes; the laws of quantum gravity then convert space into a random, probabilistic froth.

This random, probabilistic froth is the thing of which the singularity is made, and the froth is governed by the laws of quantum gravity. In the froth, space does not have any definite shape (that is, any definite curvature, or even any definite topology). Instead, space has various probabilities for this, that, or another curvature and topology. For example, inside the singularity there might be a 0.1 percent probability for the curvature and topology of space to have the form shown in Figure 13.7a, and a 0.4 percent probability for the form in Figure 13.7b, and a 0.02 percent probability for the form in Figure 13.7c, and so on. This does *not* mean that space spends 0.1 percent of its *time* in the form (a), 0.4 percent of its *time* in the form (b), and 0.02 percent of its *time* in the form (c), because *there is no such thing as time inside the singularity.* And similarly, because there is no time, it is totally meaningless to ask whether space assumes the form (b) “before” or “after” it assumes the form (c). The only meaningful question one can ask of the singularity is, “What are the probabilities that the space of which you are made has the forms (a), (b), and (c)?” And the answers will be simply 0.1, 0.4, and 0.02 percent.

Because all conceivable curvatures and topologies are permitted inside the singularity, no matter how wild, one says that the singularity is made from a probabilistic foam. John Wheeler, who first argued that this must be the nature of space when the laws of quantum gravity hold sway, has called it *quantum/foam.*